A Bayesian Network Model for Contextual versus Non-Contextual Driving Behavior Assessment

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Abstract

Driving behavior is generally considered to be one of the most important factors in crash occurrence. This paper aims to evaluate the benefits of utilizing context-relevant information in the driving behavior assessment process (i.e. contextual driving behavior assessment approach). We use a Bayesian Network (BN) model that investigates the relationships between GPS driving observations, individual driving behavior, individual driving risks, and individual crash frequency. In contrast to prior studies without context information (i.e. non-contextual approach), the data used in the BN approach is a combination of contextual features in the surrounding environment that may contribute to crash risk, such as road conditions surrounding the vehicle of interest and dynamic traffic flow information, as well as the non-contextual data such as instantaneous driving speed and the acceleration/deceleration of a vehicle. An information-aggregation mechanism is developed to aggregates massive amounts of vehicle GPS data points, kinematic events and context information into drivel-level data. With the proposed model, driving behavior risks for drivers is assessed and the relationship between contextual driving behavior and crash occurrence is established. The analysis results in the case study section show that the contextual model has significantly better performance than the non-contextual model, and that drivers who drive at a speed faster than others or much slower than the speed limit at the ramp, and with more rapid acceleration or deceleration on freeways are more likely to be involved in crash events. In addition, younger drivers, and female drivers with higher VMT are found to have higher crash risk.

Keywords: contextual driving risk analysis; Bayesian network model; information-aggregation; regression models;

1. Introduction

Driving behavior is generally considered to be one of the most important factors in crash occurrence, yet due to the stochastic nature of driving, measuring and modeling driving behavior continues to be a challenging topic today (Sagberg *et al.* 2015). In most prior studies, driving behavior data was collected from self-reported surveys, such as the Driving Behavior Questionnaire (DBQ) (Rowe *et al.* 2015), driving simulator, (Chen *et al.* 2013) or actual in-vehicle observations (Toledo and Lotan 2006). The relationship between driving behavior and crash involvement is generally assumed and studied (Guo *et al.* 2010, Paefgen *et al.* 2012, Paefgen *et al.* 2014, Sun *et al.* 2017), with the primary purposes being to distinguish the driving behavior of safe drivers versus unsafe drivers (Jun *et al.* 2011), to identify critical safety-related events (Wu *et al.* 2014), and to understand the variant in driving behavior. With these studies, individual driving behavior can be assessed and used as a representative of crash risk, and broader applications can be built upon this foundation, such as novice driver education (Simons-Morton *et al.* 2013), feedback to improve driving (Toledo *et al.* 2008, Ellison *et al.* 2015), and Pay-As-You-Drive (PAYD) insurance programs (Ferreira and Minike 2010, Paefgen *et al.* 2014).

Depending on whether driving information is context relevant, driving behavior can be contextual or non-contextual. Generally speaking, the motion of a vehicle, described by vehicle kinematic parameters like speed and acceleration/deceleration, is relied on to measure driving behavior (Af Wåhlberg 2008, Paefgen *et al.* 2011, Ellison *et al.* 2015) which could be referred to as a non-contextual driving behavior approach, as it solely applies vehicle information, but does not consider the surrounding environment of the driver that may contribute to crash risk such as road conditions surrounding the vehicle of interest and dynamic traffic flow information. Research utilizing non-contextual driving behavior approach has its limitation. For example, if a driver exhibits stop-and-go or abrupt accelerate/decelerate behavior, without supplemental information on traffic conditions, it is usually difficult to tell if this is simply due to the driver's behavior, or caused by heavy traffic conditions. Relative speed to the average traffic speed is another example. If a driver is observed to be consistently driving at a speed much higher than the other people around him/her, it's reasonable to suspect that most likely this driver is driving more aggressively than average drivers. Following this rationale, this paper proposes to define driving behavior by incorporating both habitual driving behavior (i.e. driving style) and instantaneous reactions under various contexts (i.e. contextual actions), and aims to evaluate the benefits of utilizing context-relevant information in the driving behavior assessment process (i.e. contextual driving behavior assessment approach), compared with prior studies without context information (i.e. non-contextual approach).

Prior research with the contextual driving behavior assessment approach is rather limited. Jun et, al. investigated whether the exposure of crash-involved drivers and driving performance were different on the basis of disaggregated analyses by facility types and trip start times through a 14-month study (Jun *et al.* 2007). In a prior study, by considering contextual driving behavior, including relative speed, which is measured by the combination of vehicle moving speed, real-time traffic speed and speed limit of the road, we verified that the contextual driving behaviors have strong relationships with the driver's crash involvement (Zhu *et al.* 2016). However, certain research questions are still yet to be answered, which leads to the research objectives of this paper.

1. Are there any benefits of utilizing contextual driving behavior assessment approach as opposed to non-contextual approach, and if so, to what extent?

2. With the challenges in information aggregation and the various data size volume for each driver, how do we extract contextual and non-contextual driving behavior from the detailed longitudinal vehicle telematics data at the GPS point level, combined with corresponding contextual and spatial-temporal features at the road segment level?
3. How could we relate individual driving behavior with crash involvement frequency when data is aggregated at the driver level, and explain the relationship?

To answer the questions above, a hierarchical Bayesian Network (BN) model is proposed as the methodology foundation with the vehicle trajectory data and corresponding spatial-temporal contextual data as input. The Bayesian method is effective in handling massive trajectory data, and an information-aggregation mechanism and regression models are built to fulfill the second and third research objectives. The advantages of the proposed information-aggregation mechanism lies in its flexibility in the data aggregation process, where massive amounts of GPS level vehicle kinematics context data and drivel-level data can have their own statistical property, and its capability in handling various volumes of data for each driver. With the regression model embedded in the BN structure, the relationship between driving behavior and crash risk can be investigated and explained, and a close estimate of the driver's crash risk can be obtained. With the proposed framework, two BN models are further presented to quantitatively evaluate the benefit of contextual driving behavior, by comparing the first model with only vehicle kinematics events but without contextual features, and the second model with both vehicle kinematics events and context-relevant information. The contextual information used in this study includes roadway type, which indicates various road facilities types such as freeway, arterial, ramp, etc., and speed related information including speed limit and average traffic speed information. The detailed data utilized and methodology will be described in section 3.

With this modeling framework, an understanding of the contextual driving behaviors and their relationships with crash involvement can be achieved. The outcome of the research can be used to actively advise the drivers, and encourage safe driving behaviors under specific scenarios to reduce the incidences of traffic crashes. In the sections below, literature focused on driving behavior assessment from vehicle kinematics will firstly be reviewed. Next, the proposed hierarchical Bayesian Network model with three main components will be presented. The following case study section describes the data, compares the results of two BN models, and discusses the BN model for contextual driving behavior assessment. The research contributions and limitations will be summarized at the end of this paper.

2. Literature review

The studies on driving behavior assessment have not settled on a common framework due to the varieties of data collection. For example, in the 100-car naturalistic driving study in VTTI (Neale *et al.* 2002), four levels of driving events--crash, near-crash, incidents, and baseline-- are classified from the video by trained video reductionists. In the study by the University of Michigan Transportation Research Institute (UMTRI) (Koziol *et al.* 1999), longitudinal data from videos and sensors, such as time headway, velocity, acceleration, braking force and response time, are used to identify driving errors and categorize them as near-crash, hazard present, or no hazard. This section will systematically review prior studies from four aspects including indicators, crash risks, methods, and contextual driving behaviors.

2.1. Indicators

The data utilized for driving behavior analysis typically comes from the Global Position System (GPS), in-vehicle data recorder (IVDR), instrumented vehicle, motion sensors, smartphones or other location-aware devices. As a result there is no uniform format of the data in terms of resolution, direct or derived observation, or data quality. Acceleration is usually considered as the top indicator to assess driving behavior, and braking events are commonly considered to be unsafe and highly related to crash occurrences. The rate of hard breaks or hard starts is easy to extract to the individual level from the vehicle motion and the total exposure as mileage of travel. In a study by Simons-Morton *et al.* (2013), the hard break and hard start events are referred to as "elevated gravitational force" events, and can be used to predict at-fault crashes for teenage drivers. Bagdadi (2013) found a positive correlation between the driver's crash involvement and safety critical braking events, which are suggested to be used for assessing high risk drivers. In the study by Wang *et al.* (2015), braking is measured from maximum deceleration, average deceleration, and a percentage of reduction in vehicle kinetic energy to present driving risks. The speed profile of a driver is another indicator of driving behavior, which is used to assess whether a driver is aggressive. The variation in the speed is also a signal of unsafe driving behavior, as cautious drivers are expected to maintain a moderate speed while driving when compared with aggressive drivers. Ayuso *et al.* (2014, 2016) found that the portion of speeding distance reduces the time of the first crash for young drivers under age thirty, reflecting a higher risk of crashes with excessive speed. This is especially true for male drivers. In addition to speed and acceleration that reveal longitudinal vehicle motion, some other indicators such as lateral acceleration and yaw rate (Simons-Morton *et al.* 2013) can be used to reveal the lateral vehicle motion and the steering behavior and lane changing behavior of a driver. Also, as mentioned by Eboli *et al.* (2016), combined kinematic parameters should be used if speed and acceleration are interrelated. In research such as Ellison *et al.* (2015) and Ellison *et al.* (2012), individual driving risk is profiled based on vehicle speed over the speed limit and acceleration, together with maximum, average, minimum and standard deviation of speed, acceleration and deceleration, number of sharp kinematic events, and additional indicators. Af Wåhlberg (2008) also proved that deceleration is superior to speed to a certain extent in predicting the crash involvement, and the combination of acceleration and deceleration provides a better modeling result. Overall, in most of the studies, *speed* and *acceleration* are the core indicators used for evaluating driving behavior, and real-time contextual information is not associated with the vehicle motion, which could be classified as a non-contextual driving behavior assessment.

2.2. Crash risks

To examine the correlation between driving behavior and crash risk, the data obtained should be able to represent the crash risks of the driver, such as the occurrence and frequency of crashes during the naturalistic study period (Jun *et al.* 2011, Ayuso *et al.* 2014). However, the number of crashes observed during the naturalistic driving study is usually small and insufficient to support the study analysis. Therefore, surrogate measures are generally used instead of the true number of crashes from observation. Near-crash is one of the surrogate measures utilized in assessing the crash risks and unsafe events in naturalistic driving (Klauer *et al.* 2006, Guo *et al.* 2010). Similarly, Simons-Morton *et al.* (2013) reviewed videos to determine the at-fault events of the drivers resulting in a crash or near-crash, and used the results as the dependent variable.

Another approach of crash risk is using historical crash records to represent the crash risk of a driver. For example, Jun *et al.* (2011) and Jun *et al.* (2007) compared driving behaviors between groups of drivers with and without crashes. It is found that the mileage, speed and acceleration of unsafe drivers who used to have crashes are quite different from safe drivers without any crash incidents. The crash history, mostly collected as self-reported information, can be used as the surrogate measure of crash risks based on the assumption that driving behavior is stable for general adults (as opposed to young drivers), and the current driving risk should remain stable for years for the adults measured. Drivers without a crash after years of driving, can be agreed to be labeled as safe drivers compared to those who have had crashes before. In this research, the second approach, i.e. the self-reported crash history, is utilized with the assumption stated above.

2.3. Methods

With both driving behavior and crash risks, models can be built to examine the relationship in between them. The most basic approach is to use Poisson or negative binomial models due to the count feature of the dependent variable, crash risks and near-crash events. A Multivariate logistic regression is applied in Paefgen *et al.* (2014) to prove the correlation between mileage and crash risk. Wu *et al.* (2014) used a Bayesian multivariate Poisson log-normal model to quantify the associations between near-crash and crash, largely due to its advantages in modeling at multiple levels and incorporating random effects. Simons-Morton *et al.* (2013) applied generalized estimating equations (GEE) logistic regression to investigate the occurrence of at-fault crashes or near crashes over one month with the elevated gravitational-force event rates. Survival analysis models are used by Ayuso *et al.* (2014, 2016) to identify the factors related to the duration and distance to the first crash occurrence. In the study of Wang *et al.* (2015), a k-means method and classification model are used, and a regression tree is used to unveil the relationship. Paefgen *et al.* (2013) compared several methods and concluded that the neural network model with the best classification performance and logistic regression is better suited from an actuarial standpoint. It should be noted that the method to process the data and to measure and extract driving behavior is largely dependent on the data collected, including its source and format, and the above methods, together with other various models, have been applied to assess driving behavior due to their various types of data and for research purposes. In this study, to address the needs to combine vehicle motion with the context features, and aggregation of unbalanced volume of data from the GPS-level to the driver level, we propose a new Bayesian Network approach that connects observations of vehicle motion with the number of crashes through hidden layers of driving behavior and crash risk.

2.4. Contextual driving behavior

There have been plenty of studies in driving behavior and crash risk, but limited studies dig deeper into the contextual features associated with the driving behavior. From the studies in the actuarial field, policyholder's driving context, such as the most frequent types of roads used and time of day, are considered to be related with crash involvement and used for premium adjustment (Litman 2005, Jun *et al.* 2007, Ellison *et al.* 2015). The analysis results in Jun *et al.* (2011) show that the expected correlation between higher speeds and crash risk is not the same as normal freeway driving and during peak AM hours. It is also suggested that for different roadway types and times of day, the speed metrics should be selected appropriately, instead of relying on the moving speed only. The impact of driving context is also studied in the research related to Advanced Driver Assistance Systems (ADAS) as reviewed by Rendon-

Velez *et al.* (2009). Besides the majority ADAS studies concentrating on a given context, there are a few studies also emphasizing on complex situations and investigating maneuvers for various driving context, such as Nigro *et al.* (2002). In the previous study by Zhu *et al.* (2016), the contextual driving behavior is defined and measured from the joint information of vehicle kinematics (speed and acceleration) and simultaneous context (speed limit and traffic speed). It is proved that contextual driving behaviors like relative speed and contextual-sensitive speeding have a strong relationship with crash involvement of the driver. However, the research of contextual driving behavior is still limited. A major objective of this paper is to gain a better understanding of contextual driving behavior, and evaluate its relative benefits quantitatively when compared with the non-contextual driving behavior assessment approach.

Combined together, there is still room for improvement on state-of-the-art driving behavior assessment research. In particular, technical challenges exist in terms of combining multi-source data into contextual driving observations, aggregating massive amounts of data for each driver, and the consideration of different amount of driving observations at the GPS-level from individual to individual. With this background, this paper aims to contribute from both methodology and application aspects through a Bayesian Network model, which models the relationship between GPS driving observations, individual driving behaviors, individual driving risks, and individual crash frequency. With such a proposed model, a case study is performed to quantitatively evaluate the benefits of adding contextual features to traditional vehicle kinematic observation, if any.

3. Methodology

In this paper, a hierarchical Bayesian Network is built to investigate the relationship between observed vehicle motion and a driver's historical crash involvements through the hidden layers of driving behavior and crash risk. For a given driver, the driving behavior is represented by sets of GPS observations when he/she drives, while the driving risk is reflected by his/her crash history. Besides the GPS-level observations of driving behavior, individual-level demographic information is also collected through a survey, including age, gender, approximated vehicle mileage traveled per year, etc. To analyze such multi-level, multi-source data and further assess driving behavior and crash risks of an individual driver, a Bayesian Network is adopted as a suitable model to address the technical issues described below:

1. In this study, both GPS-level data, including instantaneous driving speed, acceleration, link type, speed limit etc., and individual-level demographics, VMT, and crash history in the last 5 years is collected. Investigating the relationship between GPS-level and driver-level requires an appropriate information-aggregation scheme. Regular regression approaches have difficulties aggregating GPS-level data with individual-level data. In addition, the proposed hierarchical structure solved the over-dispersion issue by separating the accident count variance into Poisson noise and driver-level variation.
2. Each driver has a different volume of GPS-level data in the data collection process. Some individuals drive more and may have a higher volume of GPS-level data, while some others may have relatively lower GPS-level sample sizes. Such data volume variety will result in different levels of "confidence" for different individuals in the study. From this perspective, the Bayesian network handles the data volume variety perfectly, by giving each driver some "confidence level" information, which depends highly on data volume. Bayesian framework's capability of handling various data volume can be exampled by a study of Bayesian regression with missing data (Yuan and Yin 2010).

3. Compared with other heuristic algorithms such as neural networks, each node in the BN model can have its own distributions, which makes it intuitively explainable. Therefore, the driving behavior and its relation with road safety can be explained from the model results.

To explain this BN model in details, Section 3.1 briefly introduces the concept of a Bayesian Network and provides an overview of this method. Then, the structure of the BN for the purpose of this study is explained in section 3.2, with detailed model coefficients and distribution assumptions. Section 3.3 presents the computation of this model based on the Monte Carlo Marconian Chain (MCMC).

3.1. Bayesian Network Introduction

For a hierarchical system, the complexity of data structures brings challenges to the traditional regression methods. To represent this kind of hierarchical system, directed graphical models (DAGs) are a good solution. DAGs consist of nodes and arcs, which represent data, attributes, and the relationships between them. Furthermore, based on a DAG, Bayes rule can be introduced to model the nodes and arcs with probabilistic distributions. Such probabilistic directed graphical models are what we call a Bayesian Network. In a BN, the nodes represent observed data or hidden features, and every node is assumed to follow a certain distribution. The arcs of the graph represent probabilistic relationships between variables and attributes (Koller and Friedman. 2009). Typically, these relationships are primarily formulized as a conditional distribution. The posterior distributions of the hidden nodes can be learned from training the BN.

In recent years, the BN has been more popular in modeling massive amounts of data with the need for data aggregation and model flexibility (Li *et al.* 2014, Tandon *et al.* 2016), and has been adopted by different areas of study such as clinical research (Orphanou *et al.* 2014), renewable energy (Borunda *et al.* 2016), and neuroscience (Bielza and Larrañaga 2014). In the area of driving behavior analysis, Kim *et al.* (2013) built a Bayesian hierarchical regression to analyze the impact of multiple factors in assessing teenagers' driving risks. These works show the effectiveness of Bayesian methods when applied to massive trajectory data. Another application to driving behavior analysis is focused on the impact to vehicular emissions (Mudgal *et al.* 2014), which introduced a hierarchical structure to model Driver-GPS data, where a Bayesian Network model is applied for regression.

In a simple BN with three nodes (Figure 1), the nodes represent the driver's emotions, behavior and crash risk, respectively. The directed arcs show that emotion will affect an individual's behavior, and furthermore, that they jointly affect the crash risk. In BN, each node is assumed following a particular distribution, and these distributions are depending with each other based on the graphical relationship. The notation of "Risk|Emotion, Behavior," interpreted as the distribution of risk given motion and behavior, means that an individual's risk distribution relies on their current motion and behavior. Based on the observed emotion and behavior, this posterior distribution "Risk|Emotion, Behavior" is the inference of risk. If the assigned distributions of the nodes are conjugated, e.g. all normal distribution, then the inference of one node can be obtained based on a closed form solution.

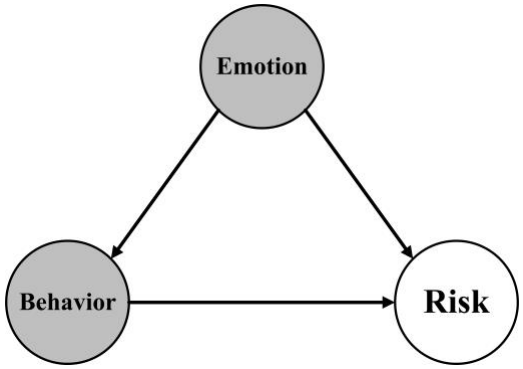


Figure 1 Simple Example of BN Model

However, for real-world problems, there may be thousands of nodes and arcs with complex distributions, which makes a closed form solution unavailable. In such case, the Monte Carlo Marconian Chain (MCMC) methods (Hastings 1970) can be applied to solve a complex BN model. Based on MCMC methods, the inference of one node will be obtained based on simulation sampling results, which is approved converging to the true distribution. By using the well-developed packages, such as JAGS for R, the inference of a BN can be easily obtained.

3.2. BN Structure

The structure of the BN designed for this study is illustrated In Figure 2. The shaded nodes are observed information, including GPS-level observations $\boldsymbol{G}_{i,j}$, drivel-level characteristics $\boldsymbol{D}_i$ and T_i, and self-reported number of crashes A_i. The unshaded nodes are the hidden attributes, including driver behavior $\boldsymbol{B}_i$, driver crash risk λ_i, and the model coefficient $\boldsymbol{\alpha}$ representing the relationship between driving behavior and crash risk. The arcs and hidden nodes link all observed information together to construct a complete BN model. From training a BN model, the posterior distribution of hidden nodes can be learned, which is the driving behavior $\boldsymbol{B}_i$, crash risk λ_i, and the model coefficient α.

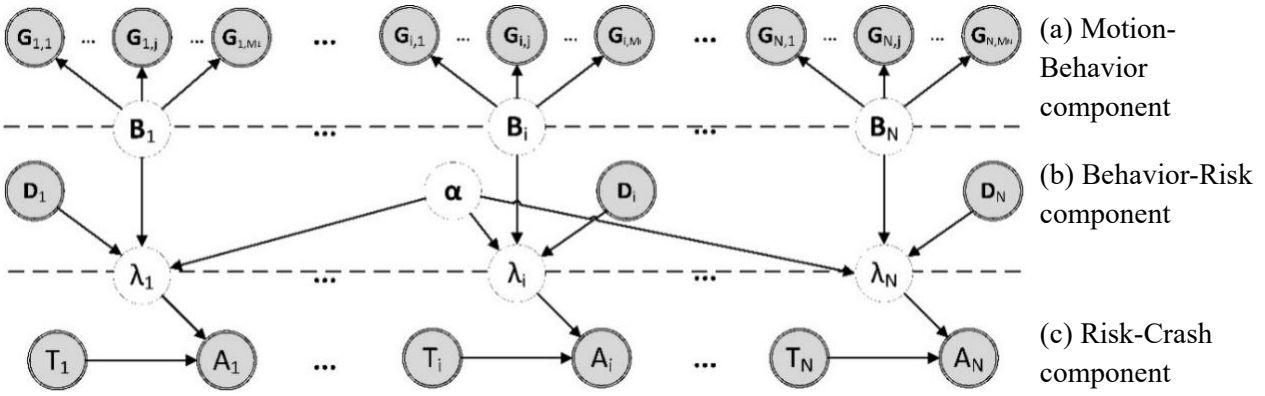


Figure 2 Bayesian Network Structure for Driving Risk Assessment

Where:

$\boldsymbol{G}_{i,j}$ is the jth GPS observation of individual i, and j = 1,2,…, M_i

M_i is the volume of GPS data for individual *i*

$\boldsymbol{B}_i$ is a set of coefficients which represent the driving behavior of individual *i*

$\boldsymbol{D}_i$ is the individual-level information for an individual *i*, including but not limited to age, gender, vehicle miles of travel (VMT) per year

λ_i is the annual driving risk of individual *i*

$\boldsymbol{\alpha}$ is the set of model coefficients to be estimated which links $\boldsymbol{D}_i$ and $\boldsymbol{B}_i$ to λ_i. It is similar to the coefficients in a regular linear regression model

T_i is driving experience (in years)

A_i is the historical number of crashes

The graphical model in Figure 2 can be divided into 3 parts:

a) Motion-behavior component, where GPS observations $\boldsymbol{G}_{i,j}$ are affected by driving behavior $\boldsymbol{B}_i$ of individual *i.* This component is the step to aggregate various volumes of GPS-level observations to driver-level behavior. Also, this component is where the differentiation of the contextual versus non-contextual model with different information could be derived, which will be further explained below.
b) Behavior-risk component, where the driving risk λ_i is constructed as a generalized linear regression model of driving behavior $\boldsymbol{B}_i$, and driver characteristics $\boldsymbol{D}_i$, because they jointly affect drivers' crash risks and how the impacts are can be learned as $\boldsymbol{\alpha}$.
c) Risk-crash component, where the reported number of crashes A_i is modeled as a Poisson distribution with time T_i and crash risk λ_i.

From the complete model, all observed information from N individuals can be aggregated to learn the posterior distribution of the hidden nodes, including individuals' driving risk λ_i and the model coefficient $\boldsymbol{\alpha}$. Based on the obtained posterior distributions, an estimation of λ_i will be used for risk assessment, and $\boldsymbol{\alpha}$ can be used for explanation and prediction purposes. This BN can be applied to both contextual behavior assessment and non-contextual behavior assessment, given two different sets of readings in $\boldsymbol{G}_{i,j}$ and two sets of formulation in component (a). For the contextual approach model, $\boldsymbol{G}_{i,j}$ includes readings of vehicle kinematics (speed and acceleration), as well as the context-relevant information the driver is facing (type of roadway and average traffic speed), which is associated with GPS-level data by the spatial and temporal information as defined in Equation (1)-(13). While in the non-contextual approach model, $\boldsymbol{G}_{i,j}$ includes only the vehicle kinematics readings, and subsequently becomes a simplified version of the contextual model with a smaller number of variables, as shown in Equation (14)-(15).

As aforementioned, all the nodes need to be assigned to some certain distributions, and all the arcs are assumed to be some conditional relationship to link the nodes. In the following subsections, more details in modeling these nodes and arcs are demonstrated for the contextual behavior models. The nodes and arcs in the motion-behavior component for the contextual model will firstly be introduced in 3.2.1. The BN construction for the non-contextual behavior model is a simplified version and discussed briefly in the end. Then the behavior-risk component and risk-crash components are described after that.

3.2.1 Motion-Behavior Component

Motion-Behavior Component is the most complicated component in this BN, where different individuals have different volumes of GPS-level data, and each GPS observation includes multiple readings such as speed, acceleration, link type etc. $\boldsymbol{G}_{i,j}$, the jth GPS observation for individual *i,,* includes three main types of information, and is defined as following:

$$\boldsymbol{G}_{i,j} = [\boldsymbol{L}_{i,j}{}^{T}, \boldsymbol{V}_{i,j}{}^{T}, \boldsymbol{\delta}_{i,j}{}^{T}]^{T} \tag{1}$$

Where

$\boldsymbol{L}_{i,j}$ is the link type information which indicates the type of link in the current position of the jth GPS observation for individual *i*

$\boldsymbol{V}_{i,j}$ is the speed-related information in the jth GPS observation for individual *i*

$\boldsymbol{\delta}_{i,j}$ is the acceleration information in the jth GPS observation for individual *i*

At first, $\boldsymbol{L}_{i,j}{}^{T}$, the link type indicator, which is a contextual feature of the GPS observation, is defined as a vector with binary variables indicating what kinds of link the driver i is driving on at the moment of the jth GPS observation:

$$\boldsymbol{L}_{i,j} = [L_{i,j}^{1}, L_{i,j}^{2}, \dots, L_{i,j}^{k}, \dots, L_{i,j}^{6}]^{T} \tag{2}$$

In this definition, $L_{i,j}^{k}$ is the indicator for link type k, k=1,2,…6. In total there are 6 types of links in this study: freeway, ramp, arterial, highway, minor road, and HOV link. $L_{i,j}^{k}$ is a binary variable, and 1 denotes that it is observed on link type k, and 0 denotes that it is not on link type k. Each observation can only belong to one link type, thus $L_{i,j}^{k}$ is constrained by the equation $\sum_{k=1}^{6} L_{i,j}^{k} = 1$.

For link type indicator vector $\boldsymbol{L}_{i,j}$, the most commonly used distribution is this categorical distribution:

$$\boldsymbol{L}_{i,j}|\boldsymbol{\pi}_{i} \sim Cat(\boldsymbol{\pi}_{i}) \tag{3}$$

Where $\boldsymbol{\pi}_{i} = [\pi_{i}^{1}, \pi_{i}^{2}, \dots, \pi_{i}^{k}, \dots, \pi_{i}^{6}]^{T}$ is the link type probability vector for individual i. The physical meaning of $\boldsymbol{\pi}_{\text{i}}$ can be loosely interpreted as the proportion of individual i driving on 6 link types.

Second, for the speed information $\boldsymbol{V}_{i,j}$, there are three types of readings, including vehicle speed, speed limit and estimated link average speed. Among them, vehicle speed is obtained directly from the GPS device. Speed limit and estimated link average speed are the contextual features for the $\boldsymbol{G}_{i,j}$, obtained from the traffic network system by matching the Unix time stamp and GPS location. In this paper, these three raw speed readings are concluded in two values in $\boldsymbol{V}_{i,j}$ defined as:

$$\boldsymbol{V}_{i,j} = [V_{i,j}^{s}, V_{i,j}^{r}]^{T} \tag{4}$$

Where

$V_{i,j}^{s}$ is the speeding value, which is defined by the vehicle speed minus the speed limit, indicating whether and how much the individual drove over the speed limit.

$V_{i,j}^r$ is the relative speed value, which is defined by vehicle speed minus estimated link average speed, indicating how much faster/slower the individual drove compared to others.

It should be mentioned that low speed readings (defined as vehicle speed lower than 5 mph) at the beginning and ending parts of the trip were removed from the database. Such readings are usually obtained when an individual is yet to start the trip or has stopped in parking lots or other similar instances. Such data is meaningless to assess individuals' speed related driving behavior, but will bring noise to the model.

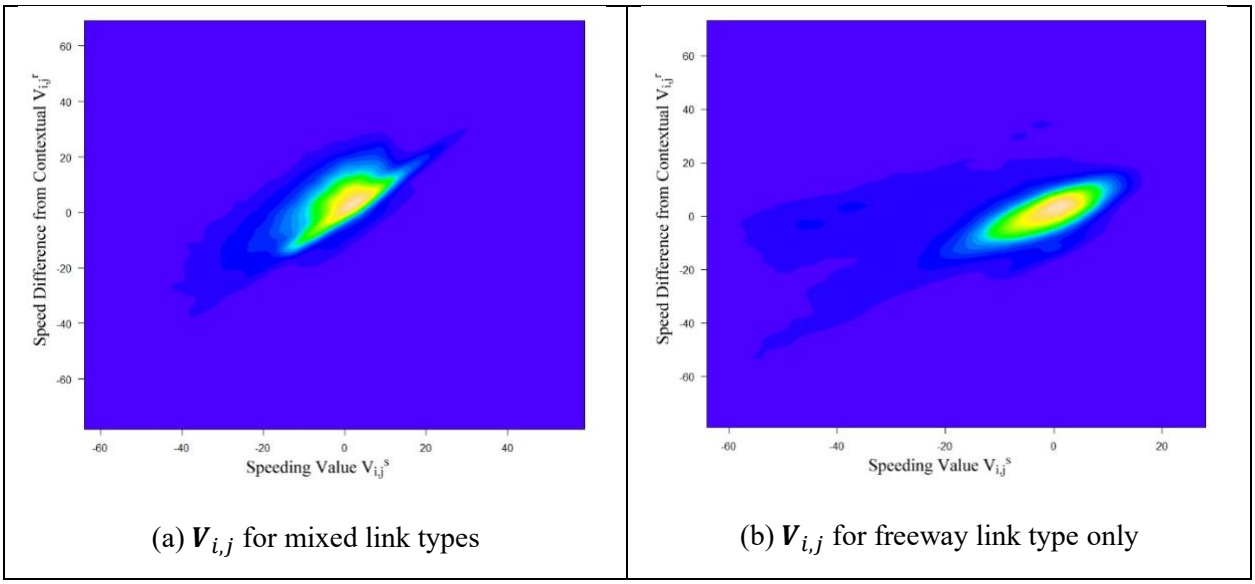
Under the contextual BN model, drivers' performance might be heterogeneous on different link types, e.g. on ramps, drivers may hit breaks more often. Such suspicion is analyzed and verified by the heterogeneity of $\boldsymbol{V}_{i,j}$ in Figure 3, where (a) shows the kernel density of $V_{i,j}$ with mixed link types and the density shows significant heterogeneity. On the other hand, the density of $V_{i,j}$ for freeway links is only demonstrated in (b), which appears to be more homogenous. Thus, the distributions of speed and acceleration information are assumed to be conditional on the given link type, and the speed related information $\boldsymbol{V}_{i,j}$ is assumed to follow a mixture of a normal distribution instead of a traditional normal distribution due to the heterogeneity in the raw data.

$$\boldsymbol{V}_{i,j}|\boldsymbol{L}_{i,j}, \boldsymbol{\mu}_{i,k} \sim \sum_{k=1}^{6} L_{i,j}^{k}\, Normal(\boldsymbol{\mu}_{i,k}, \boldsymbol{\Sigma}_k) \tag{5}$$

Where

$\boldsymbol{\mu}_{i,k}$ is the mean speed value for individual i on link type k. It also includes two elements: the mean speeding value and mean relative speed value.

$\boldsymbol{\Sigma}_k$ is the 2-by-2 covariance matrix of speed related values on link type k. Here, for different individuals, the difference of the covariance matrix is minor. For simplification purposes, the covariance matrix is assumed to be identical for all individuals in this paper.



(a) $\boldsymbol{V}_{i,j}$ for mixed link types

(b) $\boldsymbol{V}_{i,j}$ for freeway link type only

Figure 3 Illustration of data heterogeneity

Next, the acceleration information $\boldsymbol{\delta}_{i,j}$ contains two readings:

$$\boldsymbol{\delta}_{i,j} = [\eta_{i,j}, \Delta_{i,j}]^T \tag{6}$$

Where

$\eta_{i,j}$ is defined as the kinematic event indicator, which equals 0 when the vehicle is in smooth moving status, and 1 when the vehicle is accelerating or breaking.

$\Delta_{i,j}$ is the value of the vehicle's acceleration, and a negative value of $\Delta_{i,j}$ indicates braking and a positive value indicates acceleration.

The kinematic event indicator $\eta_{i,j}$ is assumed to follow a Bernoulli distribution conditional on the given link type, as commonly used for dual-choices indicator.

$$\eta_{i,j}|\boldsymbol{L}_{i,j}, \rho_{i,k} \sim \sum_{k=1}^{6} L_{i,j}^{k}\, Bernoulli(\rho_{i,k}) \tag{7}$$

Where

$\eta_{i,j}$ = 1 represents kinematic events, and 0 stands for smooth driving

$\rho_{i,k}$ is the parameter representing a probability of kinematic events for individual *i* on link type *k*

For acceleration reading $\Delta_{i,j}$, it is assumed to follow a mixture univariate normal distribution with mean value 0.

$$\Delta_{i,j}|\boldsymbol{L}_{i,j}, \eta_{i,j}, \sigma_{i,k}^2 \sim \sum_{k=1}^{6} L_{i,j}^{k}\, \eta_{i,j} Normal(0, \sigma_{i,k}^2) \tag{8}$$

Where $\sigma_{i,k}^2$ represents individual *i*'s acceleration variance on link type k. Higher variance indicates that the driver breaks or starts harder. Theoretically, the summation of all $\Delta_{i,j}$ is the accumulated speed change for individual *i*. As a trip should start from 0 speed and end with 0 speed, if the summation of $\Delta_{i,j}$ equals 0, it is assumed to be the mean value of the normal distribution.

To summarize, driving behavior for individual *i* is represented by the above distribution parameters $\boldsymbol{\pi}_i$, $\boldsymbol{\mu}_{i,k}$, $\rho_{i,k}$ and $\sigma_{i,k}^2$, and driving behavior parameters are further defined as:

$$\boldsymbol{B}_i = [\boldsymbol{\pi}_i{}^T, \boldsymbol{\mu}_{i,1}{}^T, \dots, \boldsymbol{\mu}_{i,6}{}^T, \rho_{i,1}, \dots, \rho_{i,6}, \sigma_{i,1}^2, \dots, \sigma_{i,6}^2]^T \tag{9}$$

These parameters represent individuals' driving behavior, and they will be used later for driving risk assessment. Under the Bayesian framework, all these parameters should be assigned some prior distributions, and the prior knowledge is obtained from the database containing over 10 Million trajectory data from over 2,000 drivers, and is not limited to the set of studied drivers. Prior distributions are chosen based on commonly used conjugated distributions as following:

$$\boldsymbol{\pi}_i \sim Dirichelet(\boldsymbol{\pi}_0) \tag{10}$$

$$\boldsymbol{\mu}_{i,k} \sim Normal(\boldsymbol{\mu}_{0,k}, \boldsymbol{\Sigma}_{0,k}) \tag{11}$$

$$\rho_{i,k} \sim Beta(\alpha_{0,k}, \beta_{0,k}) \tag{12}$$

$$\sigma_{i,k}^2 \sim InverseGamma(\lambda_{0,k}, \theta_{0,k}) \tag{13}$$

Where $\boldsymbol{\pi}_0$, $\boldsymbol{\mu}_{0,k}, \boldsymbol{\Sigma}_{0,k}$, $\alpha_{0,k}, \beta_{0,k}$, $\lambda_{0,k}, \theta_{0,k}$ are obtained from the mean values of all drivers' behavior parameters in the database. As $\boldsymbol{L}_{i,j}|\boldsymbol{\pi}_i$ follows a categorical distribution in equation (3), and Dirichelet distribution is furtherly chosen to model the hidden node $\boldsymbol{\pi}_i$, as it is a mostly commonly used conjugate prior for categorical distribution. Similarly, all the other prior distributions are also chosen based on conjugation relationship in Bayesian analysis.

The above description is for a contextual driving behavior model, which can also be referred to as the full model. As a comparison, the reduced model only contains non-contextual driving behavior as a simplified version, where contextual readings such as ${\boldsymbol{L}_{i,j}}^T$ and average traffic speed are not available.

$${\boldsymbol{G}_{i,j}}^s = [\, {V_{i,j}^s}^T, {\boldsymbol{\delta}_{i,j}}^T]^T \tag{14}$$

$${\boldsymbol{B}_i}^s = [\, \mu_{i,s}, \rho_{i,s}, \sigma_{i,s}^2]^T \tag{15}$$

Where

${\boldsymbol{G}_{i,j}}^s$ is the GPS-level observations containing only vehicle kinematic information $V_{i,j}^s$ and $\boldsymbol{\delta}_{i,j}$.

${\boldsymbol{B}_i}^s$ represents simplified non-contextual driving behavior,

$\mu_{i,s}$ is the average non-contextual speed for individual *i*,

$\rho_{i,s}$ is the probability of kinematic events for individual *i* under non-contextual model, and

$\sigma_{i,s}^2$ is the covariance of acceleration of individual *i*.

3.2.2 Behavior-Risk Component

In this section, a regression model is introduced to link individuals' driving behaviors with their driving risk. The aforementioned individuals' driving risk λ_i is defined as the annual crash rate. For a counting process, or specifically Poisson process, the rate should strictly be a positive number. Due to this constraint, a regular linear model is not appropriate here. Instead, a generalized linear model (GLM) is applied to link driving behavior below,

$$\lambda_i = \exp(\alpha_0 + {\boldsymbol{\alpha}_1}^T \boldsymbol{B}_i + {\boldsymbol{\alpha}_2}^T \boldsymbol{D}_i) \tag{16}$$

Where

$\boldsymbol{B}_i$ is the driving behavior parameters as defined in 3.2.1

$\boldsymbol{D}_i$ is the personal profile information including an individual *i*'s age, gender, and vehicle mileage travelled per year (VMT)

$\boldsymbol{\alpha_0}$, $\boldsymbol{\alpha_1}$, and $\boldsymbol{\alpha}_2$ are the coefficients for this GLM. The number of coefficients depend on the dimension of $\boldsymbol{B}_i$ which is defined in Equations (9) and (16), and dimension of $\boldsymbol{D}_i$ which includes 3 variables as Table 1. In this study, there are 34 coefficients in the contextual BN model, and 7 in the non-contextual BN model. To be mentioned here is that unlike traditional regression, $\boldsymbol{B}_i$ and $\boldsymbol{D}_i$ are all random

variables here instead of fixed value. Thus although individual's speed variation is not directly existed in $\boldsymbol{B}_i$, but the speed variation information will be represented in the covariance of $\boldsymbol{B}_i$.

This model is the same as a link function for Poisson regression in GLM. All the parameters in $\boldsymbol{B}_i$ and $\boldsymbol{D}_i$ are productively joint based on an exponential form. The productive form is popular in similar studies about driving profiles (Ellison *et al.* 2015), but more importantly, it also guarantees the positivity of λ_i. Similarly to the previous section, the regression parameters are required to have prior distributions. The difference here is that for model coefficients, we have no information available about the impact of parameters in $\boldsymbol{B}_i$ and $\boldsymbol{D}_i$, and thus a zero-information prior is selected:

$$\boldsymbol{\alpha} = [\alpha_0, \boldsymbol{\alpha_1}^T, \boldsymbol{\alpha_2}^T]^T \sim Normal([-2.3, \mathbf{0}, \mathbf{0}], \boldsymbol{diag}(1)) \tag{17}$$

In this prior distribution, $\boldsymbol{\alpha_1}^T$ and $\boldsymbol{\alpha_2}^T$ are assumed to have 0 mean values, which means that driving behavior and individual demographics have no impact on driving risk in prior knowledge. While α_0 having a mean value of -2.3 is from prior knowledge, because the average risk rate in US is about 0.1 vehicle crash per year, and we have exp(-2.3) = 0.1 (Toups 2011).

3.2.3 Risk-Crash Component

In this BN model, an individual's crash is assumed to be affected by years of driving T_i and annual crash rate λ_i. Typically, this kind of event counting process can be considered a heterogeneous Poisson process. Based on this intuition, a Poisson distribution is assumed in this paper to represent the relationship between λ_i, T_i and A_i:

$$A_i|T_i, \lambda_i \sim Poisson(\lambda_i T_i) \tag{18}$$

To further validate this distribution assumption, the crash count of all sampled individuals is shown in Figure 4, where the X axis is the crash count and the Y axis is the frequency or probability. The red solid curve shows the true crash frequency and the dashed black curve is the fitted Poisson probability density function. It can be observed that the Poisson distribution virtually fits the crash count very well. This figure shows some over-dispersion issue for Poisson model due to heterogeneity of accident rate between drivers. This over-dispersion issue is solved by our hierarchical structure where the total variance has been modeled by two parts as:

$$Var(A_i) = E[Var(A_i|T_i, \lambda_i)] + Var[E(A_i|T_i, \lambda_i)\} = E(\lambda_i T_i) + Var(\lambda_i T_i) \tag{19}$$

Traditional regression model assumes the homogeneity of λ_i, thus $Var(\lambda_i T_i)$ is assumed as 0. However, if there exist heterogeneity of λ_i, $Var(\lambda_i T_i) > 0$ and $Var(A_i) > E(\lambda_i T_i)$, and therefore the over-dispersion issue will occur. In our hierarchical model, $Var(\lambda_i T_i)$ is not assumed as 0, instead the variance of λ_i is modeled by Equation (16), thus the over-dispersion issue is solved.

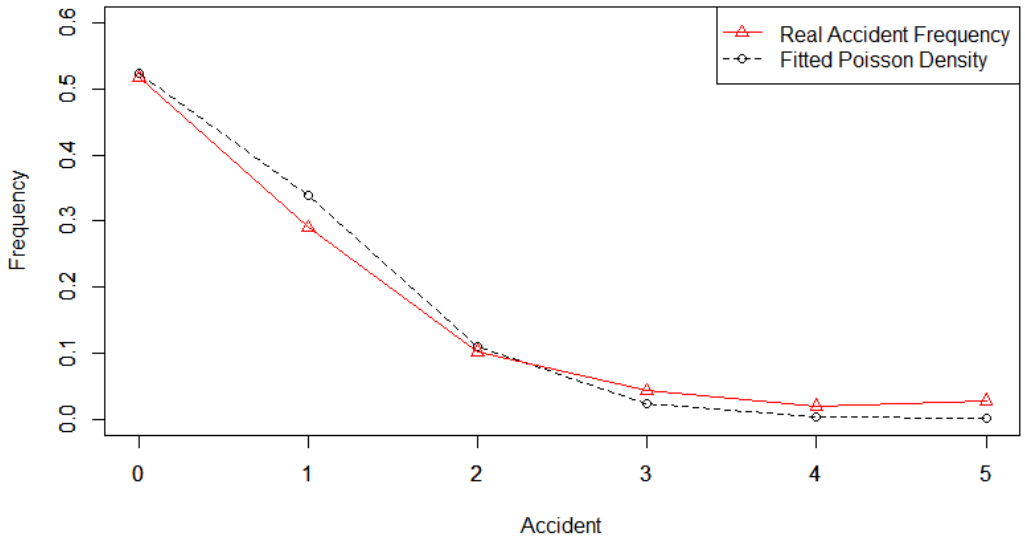


Figure 4 Poisson distribution validation

3.3 Model Computation

Typically for a BN model, the Monte Carlo Marconian Chain (MCMC) is applied to learn the posterior distributions if the prior distribution is not conjugate. However, for such a complex data structure as in this paper, directly applying MCMC will make the computational load unaffordable. Fortunately, there is a d-separation in the graph model which can help to simplify the computation (Koller and Friedman. 2009). D-separation is a special graphic structure that indicates a conditional independency relationship between two nodes. From Figure 2, node $\boldsymbol{G}_{i,j}$ and other nodes have tail to tail joints on $\boldsymbol{B}_i$, and based on this d-separation structure, we have:

$$\boldsymbol{G}_{i,j} \perp \boldsymbol{\alpha}, \lambda_i, T_i, A_i | \boldsymbol{B}_i \tag{20}$$

This independent relationship allows the information from $\boldsymbol{G}_{i,j}$ to be firstly aggregated to $\boldsymbol{B}_i$, then the rest of the learning can be achieved using the MCMC approach. Based on the distribution described in section 3.2, $\boldsymbol{L}_{i,j}$ follows a Categorical distribution with parameter $\boldsymbol{\pi}_i$, and this parameter has a Dirichelet prior distribution. Based on the Bayes rule, these two distributions have a conjugation property that makes the posterior distribution of $\boldsymbol{\pi}_i$ follow a Dirichelet distribution as well:

$$\boldsymbol{\pi}_i | \boldsymbol{L}_{i,j} \sim Dirichlet\left(\boldsymbol{\pi}_0 + \sum_{j=1}^{M_i} \boldsymbol{L}_{i,j}\right) \tag{21}$$

Similarly, other data-parameter pairs also have similar conjugation properties, and therefore the posterior distributions for other parameters also have closed forms (Koller and Friedman. 2009). After a posterior distribution of $\boldsymbol{B}_i$ is obtained by conjugate distributions, the Metropolis-Hastings (MH) Sampling method is employed to learn the posterior distribution of $\boldsymbol{\alpha}$ and λ_i (Hastings 1970).

4. Case Study

A case study is conducted to illustrate the training procedure of BN model and demonstrate its effectiveness for driving risk assessment. In addition to the trajectory data from 307,204 finished trips and

2,611,838 miles traveled, we also collect crash history data from 521 drivers who finished at least 10 trips through a designed survey. For comparison purposes, two BN models, as described in section 3, are built in this section. The first model is denoted as a contextual model, which utilized all the available information, including contextual data, such as link type and estimated link speed, while the second model assumes that contextual information is not available, which is denoted as a non-contextual model. Subsection 4.1 briefly introduces the raw data and subsection 4.2 shows the results from the BN models. Discussion on risk control, model advantage and disadvantage is presented in 4.3.

4.1. Data Description

As mentioned previously, each individuals' information has two levels: GPS-level data and Individual-level data. GPS-level data is obtained from a smartphone platform which records driver's GPS information while driving. It also includes speed information, location information, and acceleration information. A map-matching program is applied to process the raw GPS data to link the GPS coordinates with specific links in the roadway network and corresponding geometry features. GPS-level data are further filtered based on GPS quality and other criterions. For example, GPS trajectories with zero speed, zero acceleration, indicating the vehicle in stop, are considered as not valuable to model drivers' behavior and be filtered out. On the other hand, a predesigned survey is distributed to these drivers and it collects their individual-level data, including driving history, crash history, and profile information. A brief description of the raw data is displayed in Table 1.

Table 1 Case Study Data Description

Notation	Definition	Mean/Category	Standard Deviation/Percentage
GPS-Level			
M_i	Number of GPS points	186117.1	203851.8
$\boldsymbol{L}_{i,j}$	Link Type	Freeway	17.94%
		Ramp	3.52%
		Artery	56.24%
		Highway	5.14%
		Minor	16.57%
		HOV lane	0.58%
$V_{i,j}^{s}$	Speeding Value	-5.4005	0.0732
$V_{i,j}^{r}$	Relative Speed	2.0097	0.0723
$\eta_{i,j}$	Kinematic Events Indicator	Kinematic Event	39.90%
		Smooth Driving	60.10%
$\Delta_{i,j}$	Acceleration	0.0127	2.0811
Individual-Level			
$\boldsymbol{D}_i$	Age	37.29	11.17
	Gender	Male	46.26%
		Female	53.74%
	Vehicle mileage travelled	9788.87	3782.37
T_i	Driving Experience (in years)	4.8503	0.6450
A_i	Crash Count	0.7639	1.0777

4.2. Model Result

Based on the BN model and MCMC approach, each individuals' driving risk λ_i can be assessed. It should be noted that in this model, there is a high degree of collinearity between the freeway and ramp for variable $\boldsymbol{L}_{i,j}$ and $\eta_{i,j}$, which will make the MCMC very slow to converge. Therefore, in the case study, the freeway proportion and ramp proportion are combined together, yet the speed values and kinematic values for two link types are still separated. The convergence is checked based on commonly used method of comparing between-chain variance. More details about MCMC convergence criterion can be found is the review for monitoring the convergence of iterative simulation (An *et al.* 1998). The training

of Bayesian network is processed based on MCMC method, with R package JAGS. The convergence is achieved in 20 hours computing on a PC desktop with intel i5 processor. The Bayesian estimator of driving risk is obtained and denoted by $\tilde{\lambda}_i$. In addition, the MLE estimator of the driving risk $\hat{\lambda}_i$ is used for model comparison purpose:

$$\hat{\lambda}_i = \frac{A_i}{T_i} \tag{22}$$

Using $\hat{\lambda}_i$ as the criterion, the results of full model and reduced model are shown in Figure 5 and Figure 6.

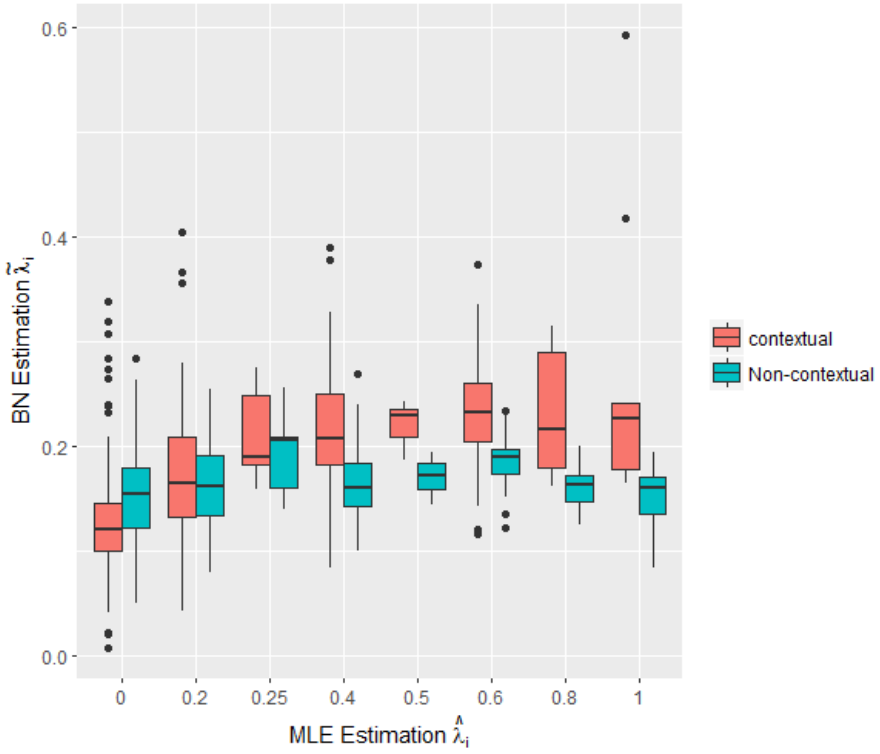


Figure 5 Bayesian Network Model Estimation Validation

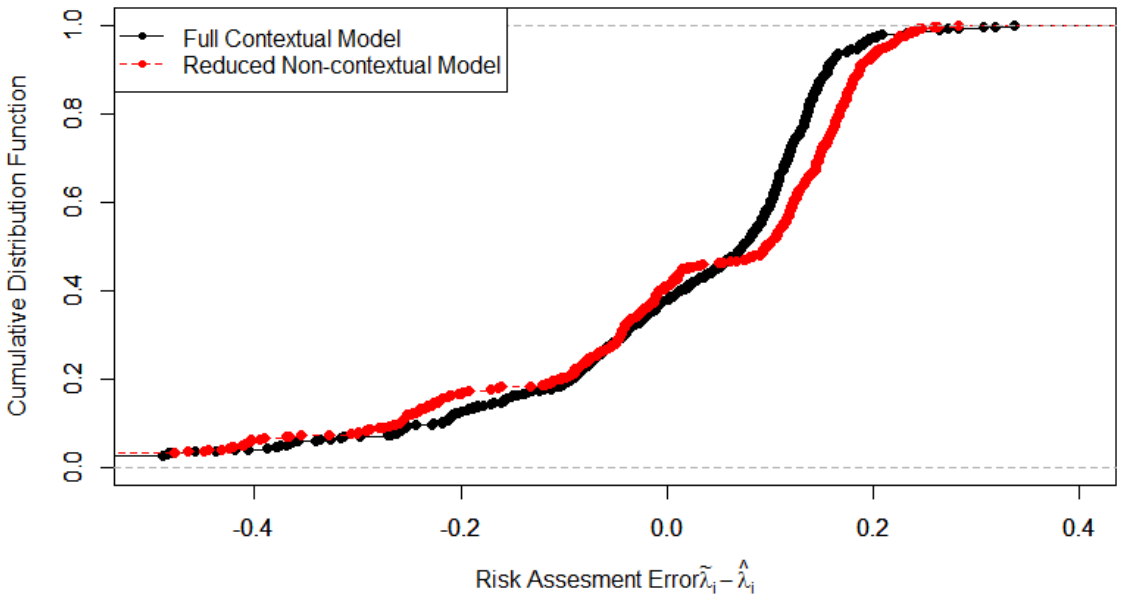


Figure 6 Cumulative Distribution Function for the accuracy of two BN models

Results from both figures show that the contextual model has significantly better performance than the non-contextual model.

1. In Figure 5, $\tilde{\lambda}_i$ (Y-Axis) from the contextual model shows a more significant correlation with $\hat{\lambda}_i$ (X-Axis), compared with the non-contextual model. For the non-contextual model, the estimated $\tilde{\lambda}_i$ appears to have no change as $\hat{\lambda}_i$ increases, while $\tilde{\lambda}_i$ from the contextual model shows a significant increasing trend along $\hat{\lambda}_i$, which is desired.

2. Numerically, the correlation between $\tilde{\lambda}_i$ and $\hat{\lambda}_i$ is 0.576 for the contextual model, which is significantly higher than the 0.143 from the non-contextual model. The pseudo-R^2 between $\tilde{\lambda}_i$ as $\hat{\lambda}_i$ also demonstrated that the contextual model is more adequate than the non-contextual model, which is 0.384 for the full model versus 0.014 for the reduced model.

3. In addition, Figure 6 shows the cumulative distribution of estimation error $\tilde{\lambda}_i - \hat{\lambda}_i$. From this plot, the full contextual model (black curve) is initially lower than the non-contextual model (red curve), but increases very fast around 0 and quickly surpasses the red curve afterwards, which shows that the modeling error from the contextual model is more concentrated at the low range level, and indicates that $\tilde{\lambda}_i$ from the full model has less errors compared with $\hat{\lambda}_i$ from non-contextual approach.

To further identify which parameters play significant roles in this model, a coefficient analysis is conducted for the full contextual driving behavior BN model and its result is demonstrated in Table 2. The parameters in the GLM model are demonstrated by Equation (16), which includes over 30 parameters. For simplification purposes, Table 2 only lists the significant parameters.

Table 2 BN Full Model Results

Parameters	Parameter Definition	Coefficient
$\rho_{i,1} + \rho_{i,2}$	Kinematic event frequency in freeway and ramp	0.3396 *
$\mu_{i,j}^{s}$	Ramp average speeding value	-0.06823 .
$\mu_{i,j}^{r}$	Ramp average relative speed	0.08184 *
$\sigma_{i,6}^{2}$	HOV lane acceleration and deceleration	-0.1556 ***
$\boldsymbol{D}_i$	Age	-0.01938 ***
	Gender	-0.1837 *
	VMT	0.07508 *
Significant Level: *** 0.001; ** 0.05; * 0.1; "." 0.2		

The coefficient analysis result shows that ramps' speed values reflect drivers' risky behavior. It's found that drivers who drive at a speed faster than others at the ramp are likely to have a higher crash rate. According to the speeding value, driving faster than the speed limit is negatively related with crash risk, indicating that a relatively lower speed under the speed limit on a ramp may be associated with a higher crash risk for the driver. The impact of these speed related driving behaviors is not significant on other types of roads, nor in the reduced non-contextual model.

The kinematic event frequency is also found to be positively significant, indicating that the more frequent a driver changes their speed on freeway, the higher crash risks they suffer from. Similar variables related with other types of road are not significant in the model. The variance of drivers' acceleration value on HOV is found to be negatively significant, indicating that a larger variance in the acceleration on HOV is related to a safer driver with less crash risk. The result is not as expected, probably due to the limited portion of HOV data. It should be noted that a larger variance of acceleration does not reveal more hard breaks and hard starts, but could also be caused by more mild acceleration and decelerations.

Lastly, drivers' individual profile information all appears to be significant. Age has the strongest significance, and young drivers are considered to have the highest rate of crash involvement. Unlike results from others, this study suggests female drivers tend to have higher crash risks than male drivers, which is probably due to the contextual BN model with control of driving behavior. The impact of VMT is positively related with crash risk, indicating that the drivers who travel more are more likely to be involved in crash risks.

4.3. Further discussion of BN model application in risk analysis and control

In addition to the above analysis that demonstrates the benefit of the BN model with contextual information when compared with a non-contextual model, another benefit under the Bayesian network is that individuals' risk assessment can be assumed to have a distribution instead of a single value, which enables more accurate risk analysis as two drivers can have the same mean estimated risk, but may have

different distributions which indicates different levels of risks. The beauty of the BN model is that such distributions can be continuously updated with improved accuracy when more data becomes available.

The number of GPS points per individual also matters. To demonstrate the effect, we firstly perform an analysis to investigate the relationship between data size and the variance of $\tilde{\lambda}_i$ for each driver in our dataset. Figure 7 shows that the variance of $\tilde{\lambda}_i$ is smaller for individuals with more GPS points, i.e., the accuracy of risk assessment is higher for drivers that have more driving data available. In other words, a higher "confidence level" could be achieved when one is a frequent user of the smartphone app.

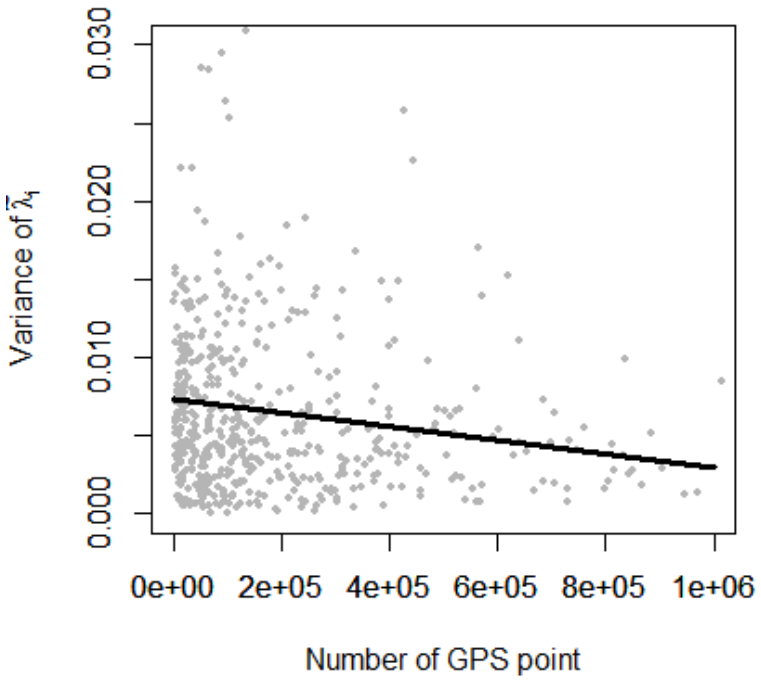


Figure 7 Estimation accuracy vs. number of GPS points

Next, we compare the risk density function of two individuals with similar profiles and identical crash histories but different data size. An illustration of risk analysis control utilizing distributions instead of mean values is demonstrated in Figure 8. The black curve stands for individual A with 111,490 GPS observations, and the red curve stands for individual B with 686,355 GPS observations. With an estimated average risk value 0.1302 for individual A and 0.1155 for individual B, their estimated mean risk is close to each other. However, if we examine their distributions, significant differences can be found between the two individuals.

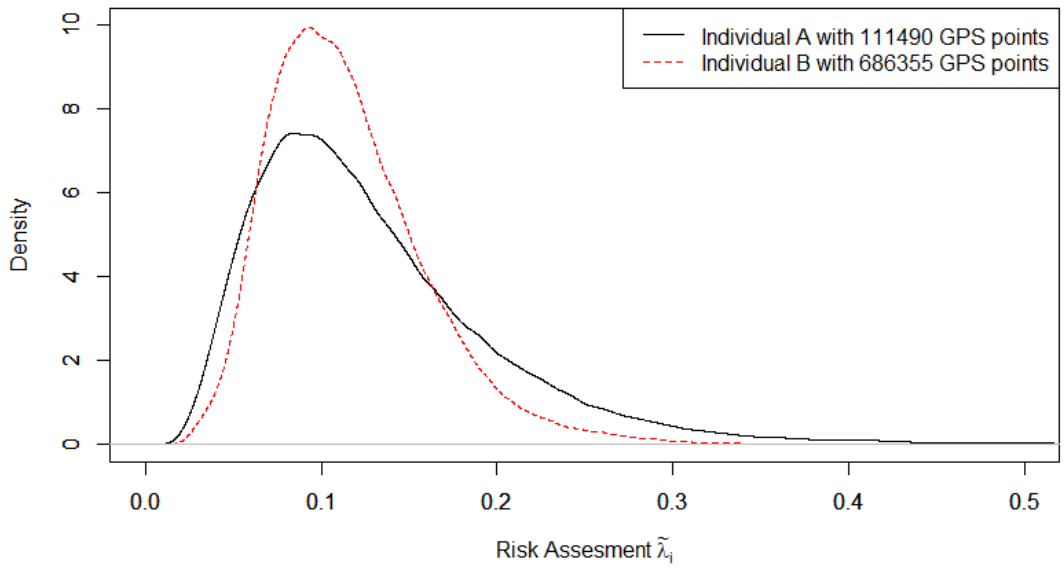


Figure 8 Illustration of Risk Estimation

As can be observed from Figure 8, the estimated risk density curve for individual B is more concentrated, and indicates a higher 'confidence' in its driving behavior when comparing to individual A. For example, if the 95% quantile is used for risk control purpose, the corresponding upper bound risk for individual A would be 0.2654, while for individual B, this value is 0.1985, which is much smaller and indicates a lower risk. In this perspective, Figure 8 demonstrates the advantage of the BN modeling approach by using a distribution instead of a mean value to describe a driver's behavior. From an application standpoint, this mechanism also indicates a practical and flexible way of implementing an online or offline driving behavior assessment model. When a new individual enters into the system, the BN model can provide a preliminary estimation based only on his/her individual-level profile data. As GPS information on this individual is collected over time, his/her driving behavior assessment can be updated regularly and/or in real-time, and the 'confidence' on his/her driving behavior assessment can be improved as the volume of GPS-data increases.

In conclusion, a BN model with a hierarchical structure is very capable of handling massive multilevel data and is appropriate for driving behavior assessment. In this paper, the BN model aggregated data from the GPS-level and individual-level to reveal the impact of both driving behavior and driver characteristics. This methodology allows for control for mis-estimated risk and yields greater confidence in the results. Compared with other heuristic algorithms such as a neural network, the nodes and arcs in the BN all have explainable physical meanings related to real word applications, which help us understand the model result. A challenge involving BN is that training a BN model relies on MCMC sampling methods. Although MCMC is proved to converge in the end, the convergence speed remains to be a problem for BN, especially in the situations where collinearity existed amongst the nodes. Variable selection is another problem for BN. Automatic removal of useless variables (nodes) might improve the efficiency of BN model training, but it is still a challenge in BN research.

5. Conclusions

In this paper, a Bayesian Network model incorporating both regression models and an information-aggregation mechanism is proposed as the methodology foundation to investigate the relationship between GPS driving observations, individual driving behavior, individual driving risks and individual crash frequency. Based on such a BN model, we demonstrate the benefits of utilizing context-relevant information in the driving behavior assessment process when comparing to prior studies without context information. In the case study section, vehicle kinematic data and simultaneous context-relevant information is collected from 521 drivers, together with their self-reported crash involvement. The modeling results with context-relevant data are proved to be much better than those without context-relevant data. The results from the contextual behavior assessment model reveal that driving at a speed faster than others on the ramp is significantly related to a higher crash risk, and a speeding value and driving faster than the speed limit are negatively related with crash risk, indicating a relatively lower speed under the speed limit on the ramps may be associated with a higher crash risk for the driver. The risky drivers are found to have more rapid acceleration and decelerations on freeway, which are defined as kinematic events frequency. Besides these driving behavior indicators, younger, female drivers with more VMT are shown to have higher annual crash risks.

With this research, a better understanding on the contextual driving behaviors and their relationship with the crash involvement is achieved. The outcome of the research can be used to assess the safety-related driving behavior for drivers, to actively advise drivers, or to encourage safer driving behaviors to reduce traffic crashes. There are still some limitations and improvements that can be looked at in future research. For example, the contextual indicators can be more comprehensive and include more variables beyond current roadway type, relative speed and traffic speed. In addition, this paper assumes drivers' driving behaviors are stable during a period of three years because of the rare observations of true crash occurrences for these drivers during the study period. This can be improved with continuous data collection from a larger sample of drivers and for a longer period to obtain sufficient observations on the real crash occurrences. Another potential direction is to improve the understanding of crash risk distribution based on the sampling of the GPS data for each person, and to better utilize this distribution to evaluate an individual's driving behavior. Furthermore, the collinearity in BN models makes the MCMC slow to converge. While we were able to proceed after manual combination of several highly correlated terms, a collinearity elimination mechanism can be developed in the future to improve the process.

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